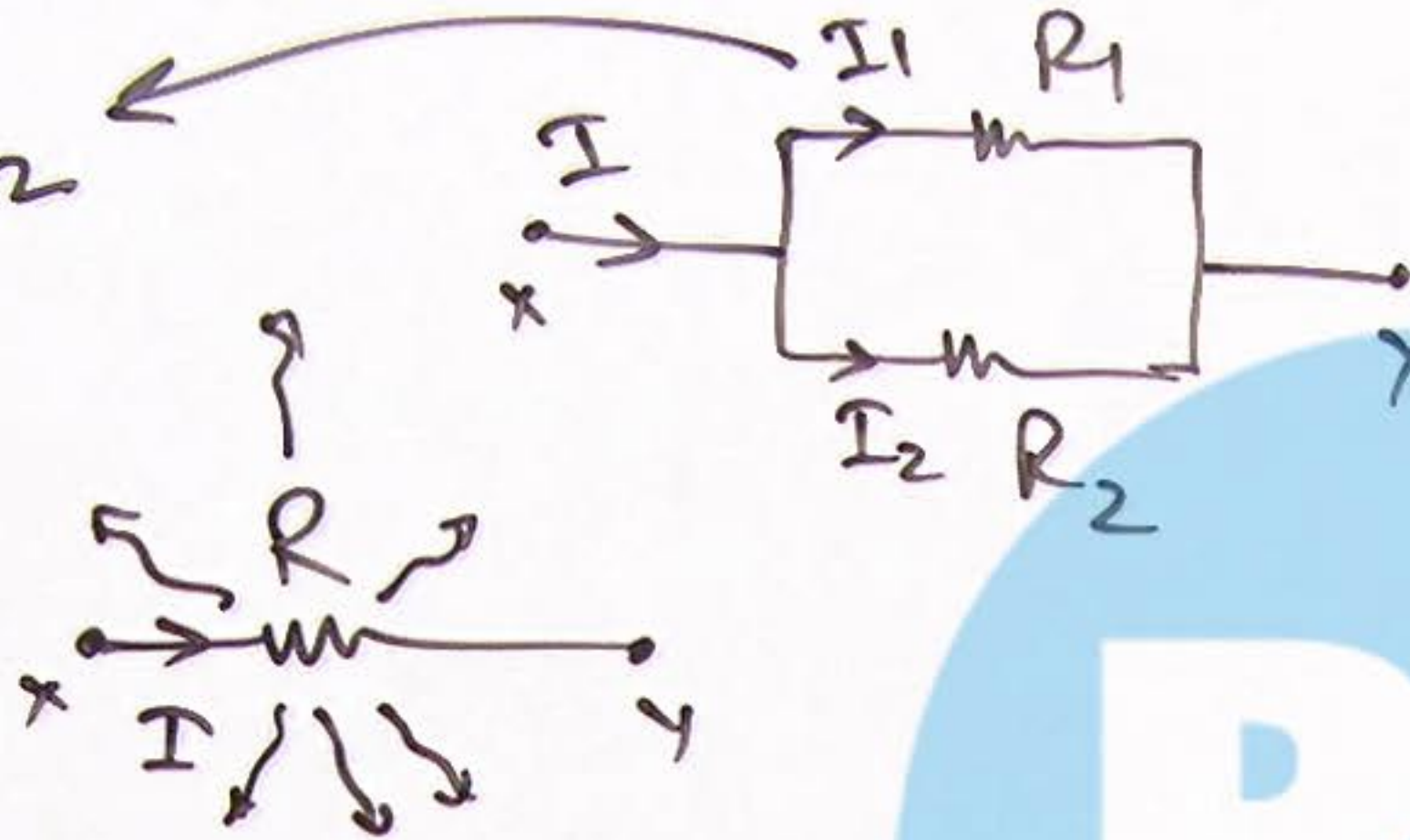


THERMAL EFFECTS, RC CIRCUITS & MEASURING INSTRUMENTS

① Thermal Effects of Current:

$$I_1 = \frac{IR_2}{R_1 + R_2}$$



Rate of heating / Thermal Power

$$V_{xy} = IR$$

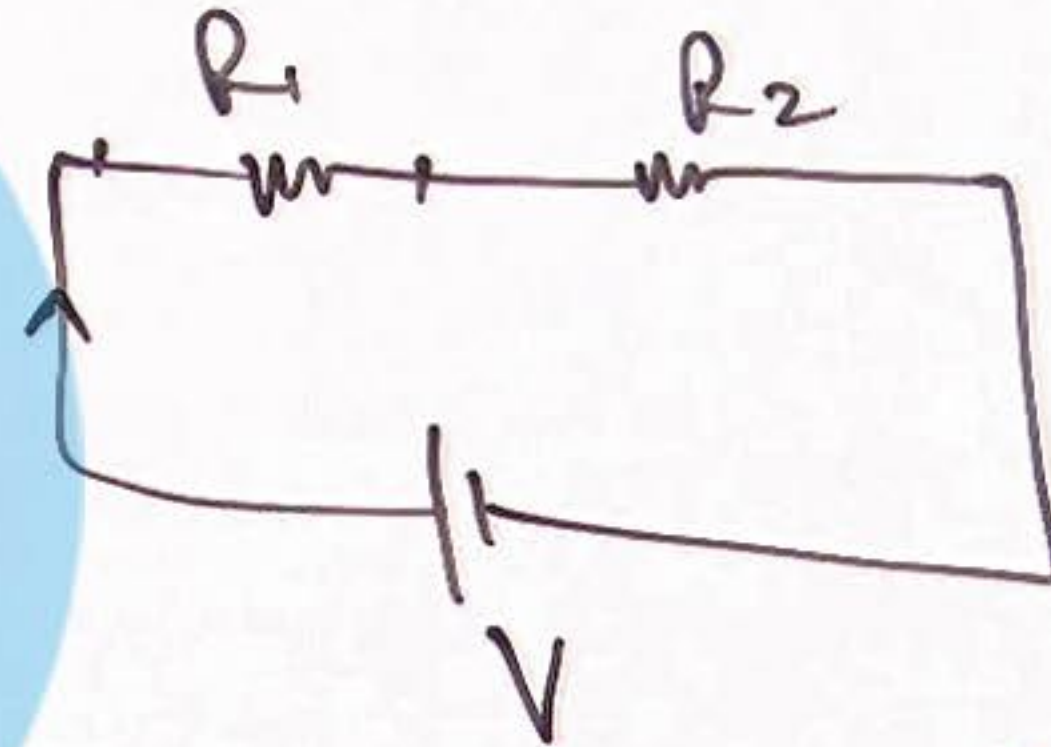
$$H = I^2 R \text{ J/s} = \frac{V_{xy}^2}{R} \text{ J/s}$$

$$P_R \neq \frac{V_{xy}^2}{R_1} = I_1^2 R_1 = \frac{I R_2 R_1}{(R_1 + R_2)^2}$$

$$V_{xy} = IR_1$$

$$P_{R_1} = \frac{V_{xy}^2}{(R_1 + R_2)}$$

$$P_{R_2} = I^2 R_2$$



$$P_R \neq \frac{V^2}{R_1}$$

$$P_{R_1} = I^2 R_1 \checkmark$$

$$= \frac{V_{xy}^2}{R_1}$$

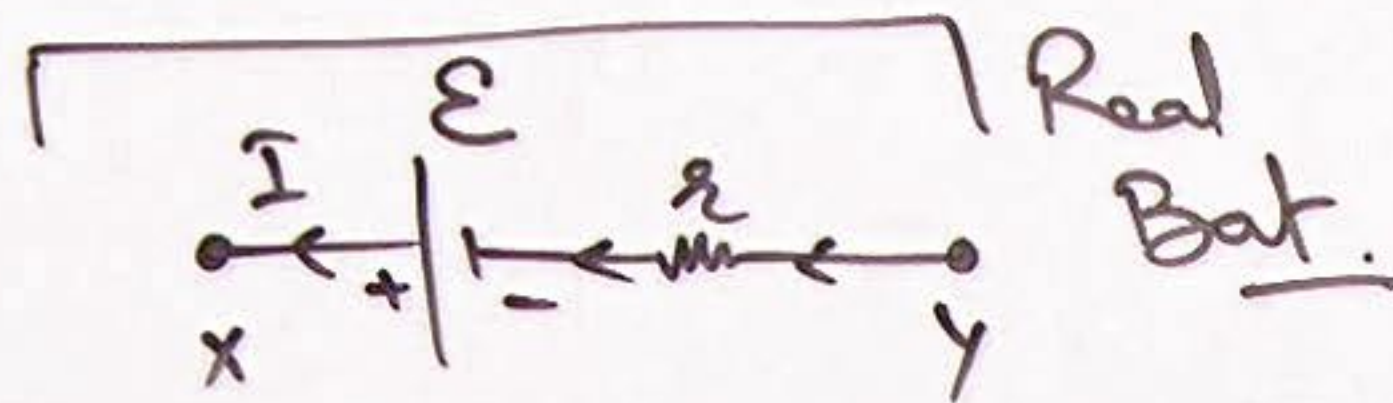
$$\frac{V}{R_1 + R_2} = I \checkmark$$

Physics Galaxy Videos

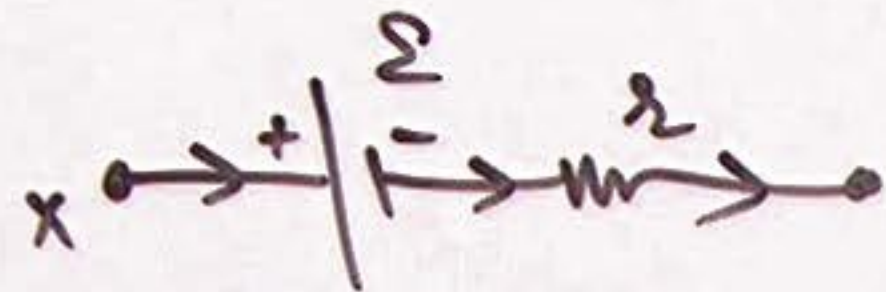
② Battery Power :

Power
Supplied by
a Battery

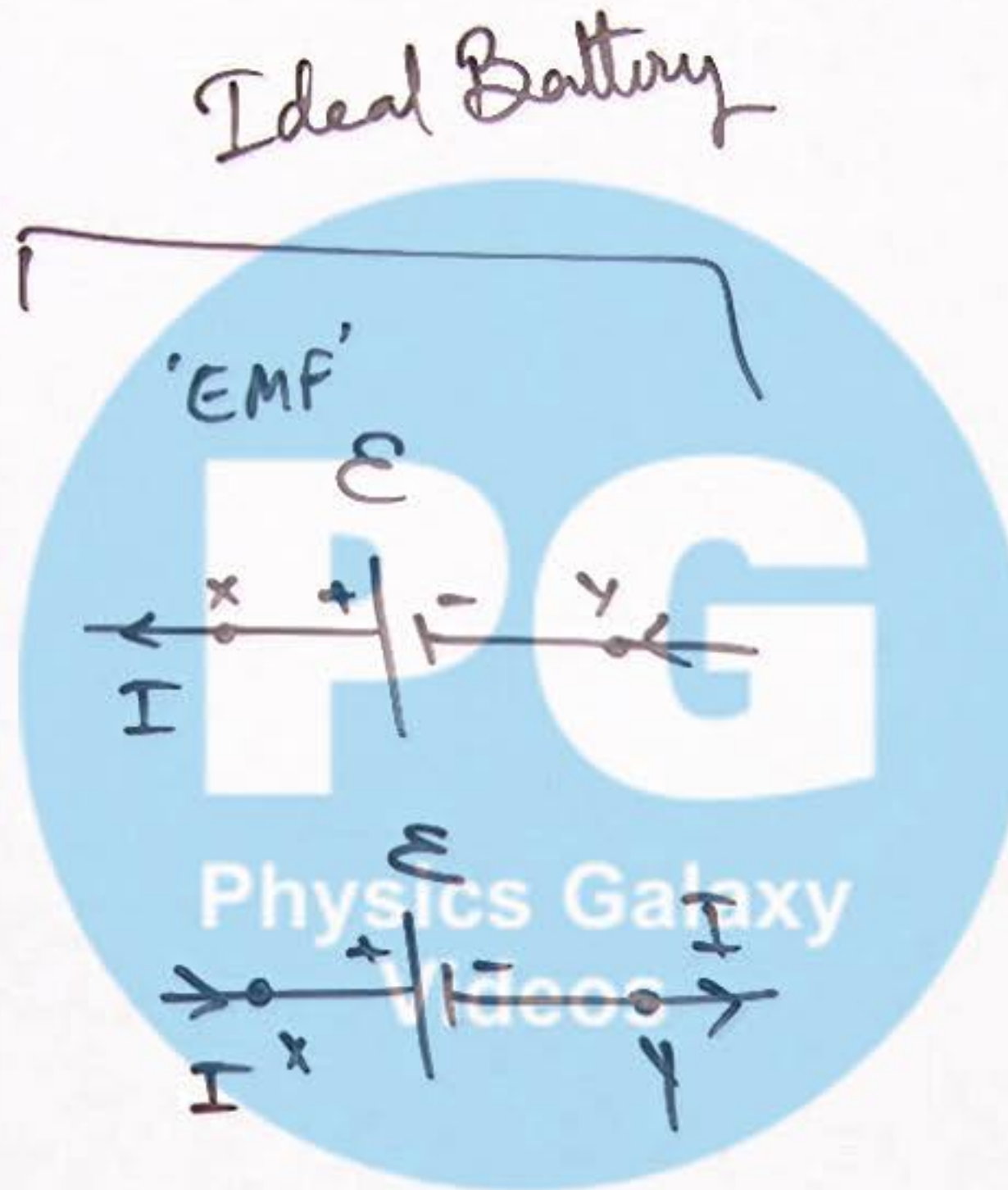
Power
absorbed by
a Battery



$$P_S = \varepsilon I - I^2 r$$



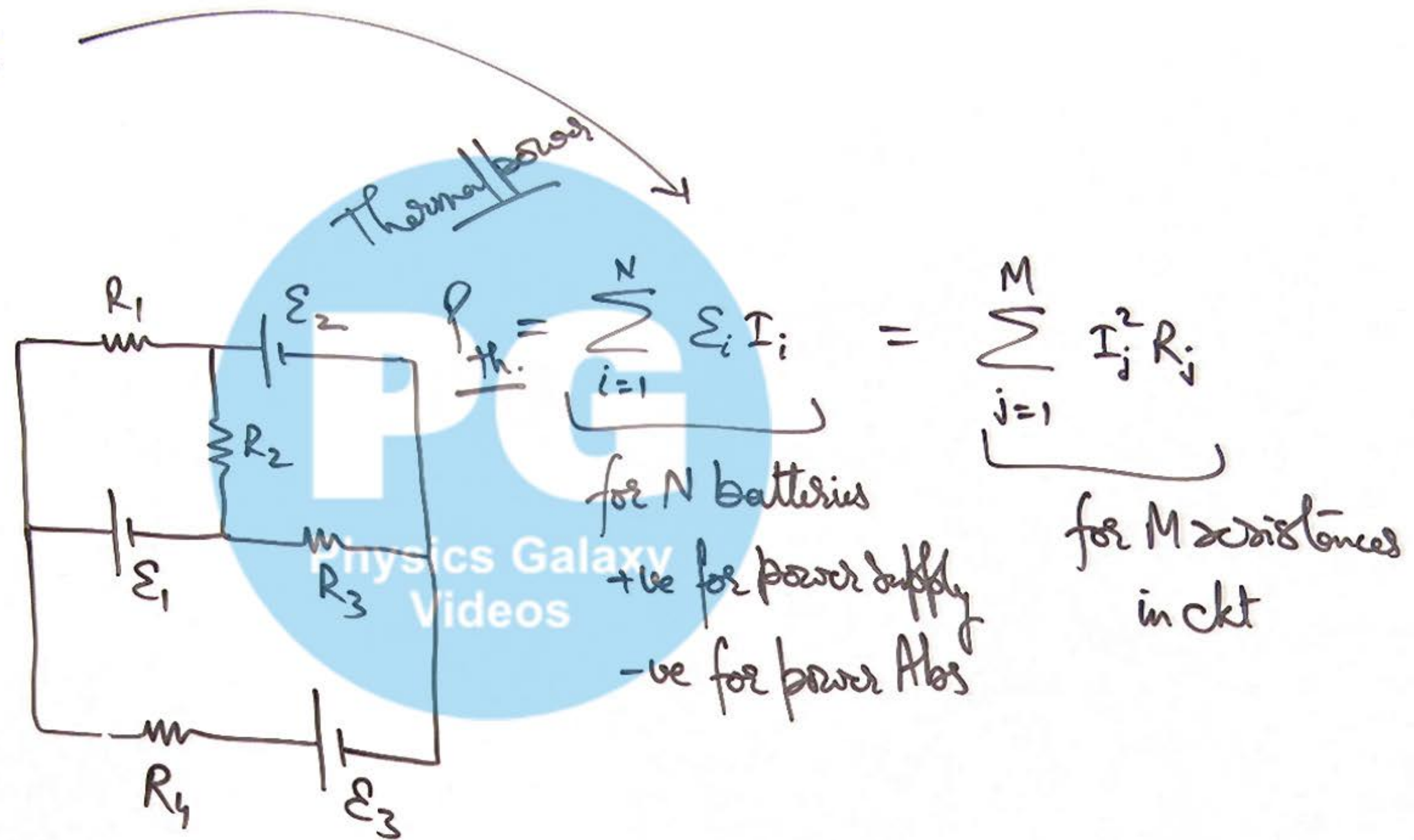
Total power consumed by bat $P_C = \varepsilon I + I^2 r$



Power supplied by bat $P_S = \varepsilon I$

Power Absorbed by bat $P_A = \varepsilon I$

③ Energy Conservation in an electric Circuit:

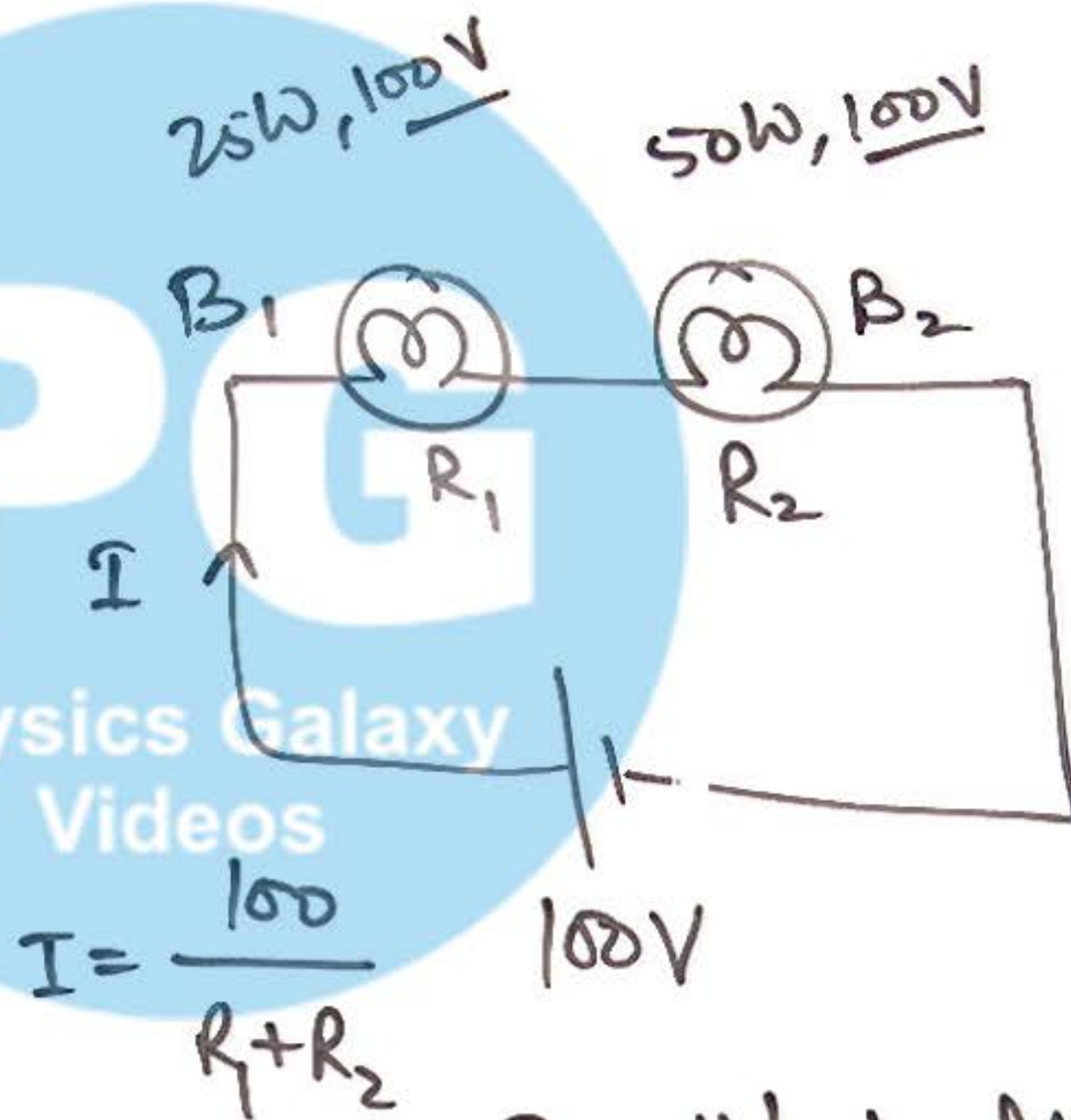


④ Brightness of Bulbs in Circuit:

Rating of Bulb. $\left[\begin{array}{l} \text{Power / Voltage} \\ 25W, 200V. \end{array} \right.$

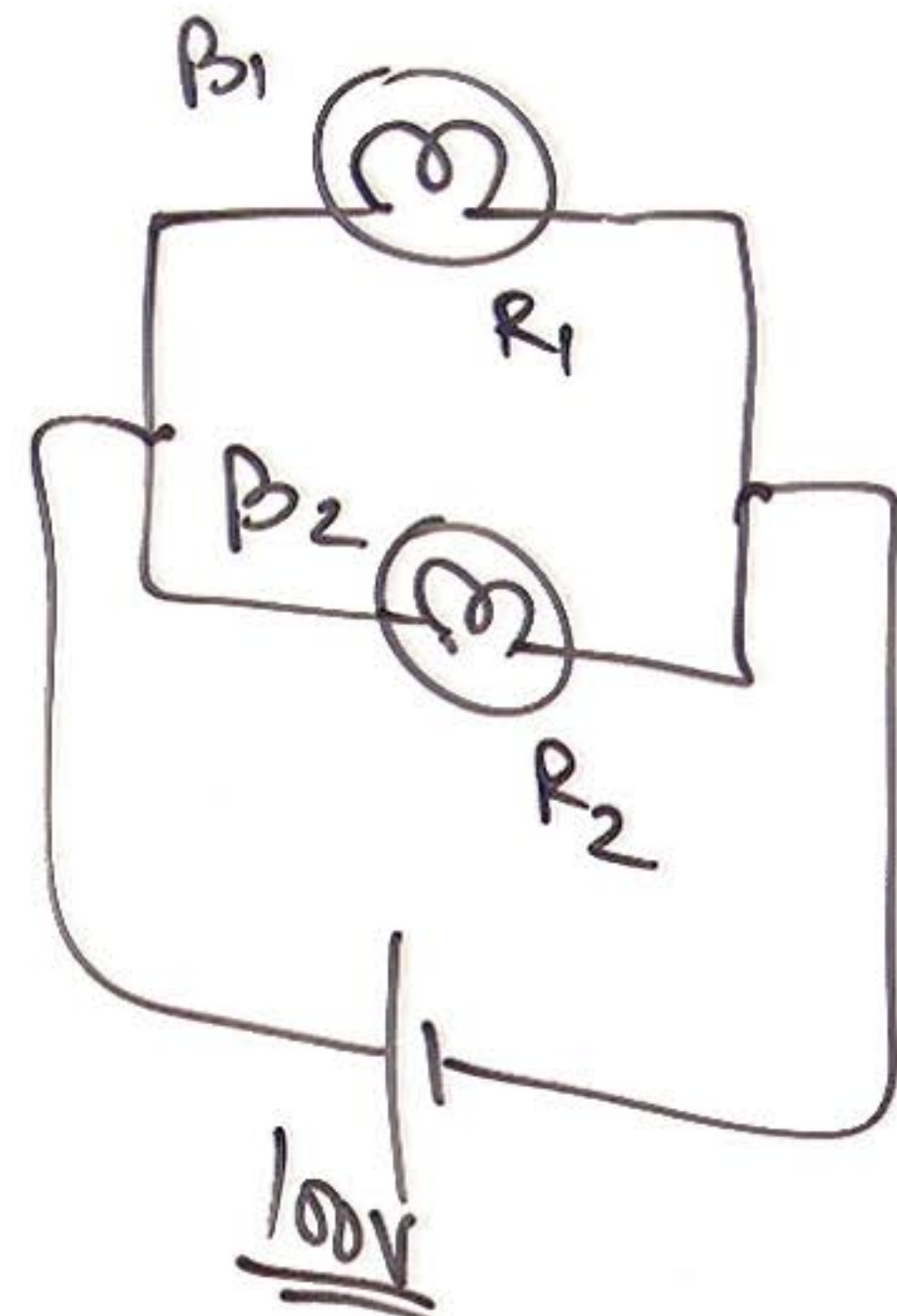
Resistance of Bulb

$$R = \frac{V^2}{P} = \frac{(200)^2}{25}$$



B_1 will be brighter

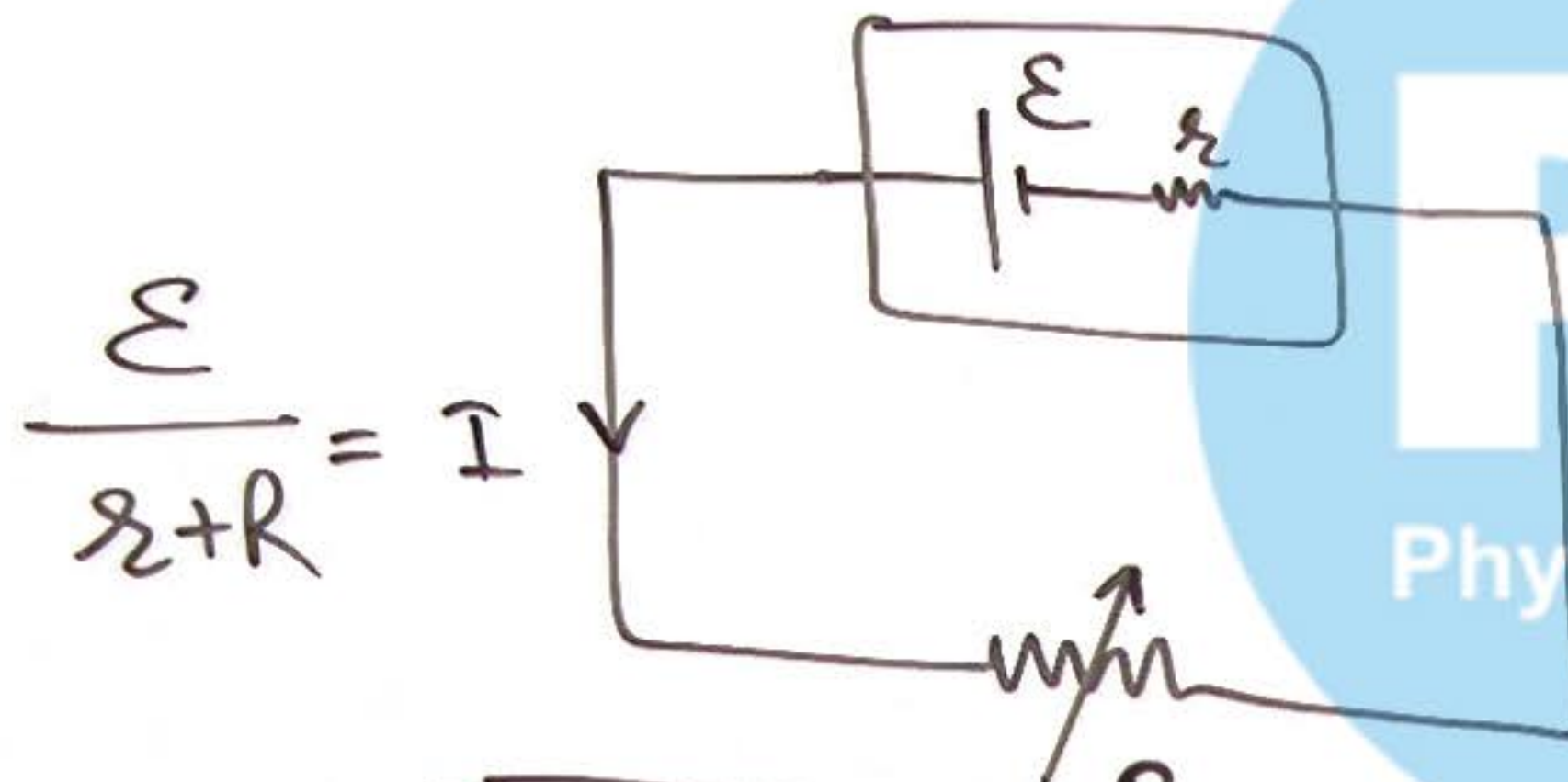
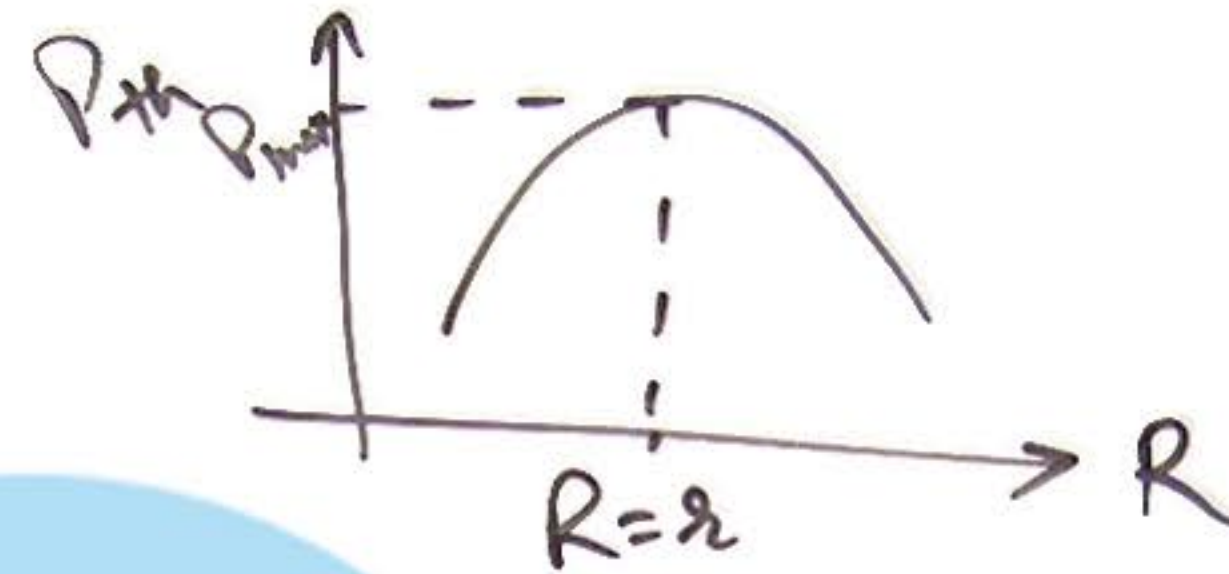
$$P = \frac{V^2}{R}$$



B_2 will be brighter

⑤ Maximum Power Transfer Theorem:

When ext res = int res
then P is max transferred



$$P_{\max} = \frac{\varepsilon^2 r}{4r^2} = \frac{\varepsilon^2}{4r} \quad \checkmark$$

Heat produced / Power transferred
to external resistor is

$$P_{th} = I^2 R = \frac{\varepsilon^2 R}{(r+R)^2}$$

for max thermal power.

$$\frac{dP_{th}}{dR} = 0 \Rightarrow R = r$$

⑥ Thevenin's Analysis :

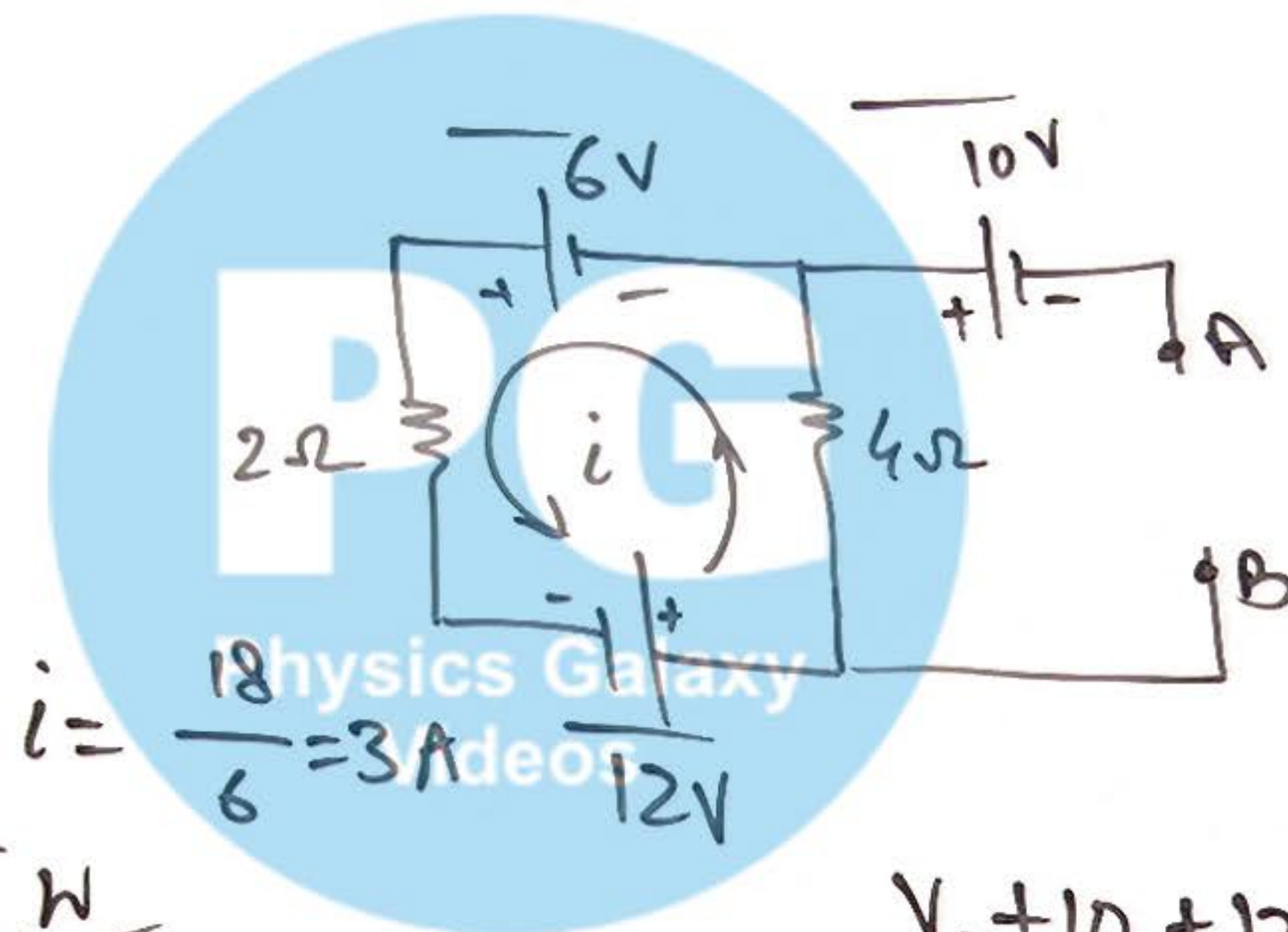
for making eq battery
for any ckt.

Power in R is max when

$$R = \frac{4}{3} \Omega$$

$$P_{\max} = \frac{\mathcal{E}^2}{4r} = \frac{(22)^2}{4 \times \frac{4}{3}} \text{ W}$$

IL: Find R for max power
in it & also find $P_{\max}$ transfer



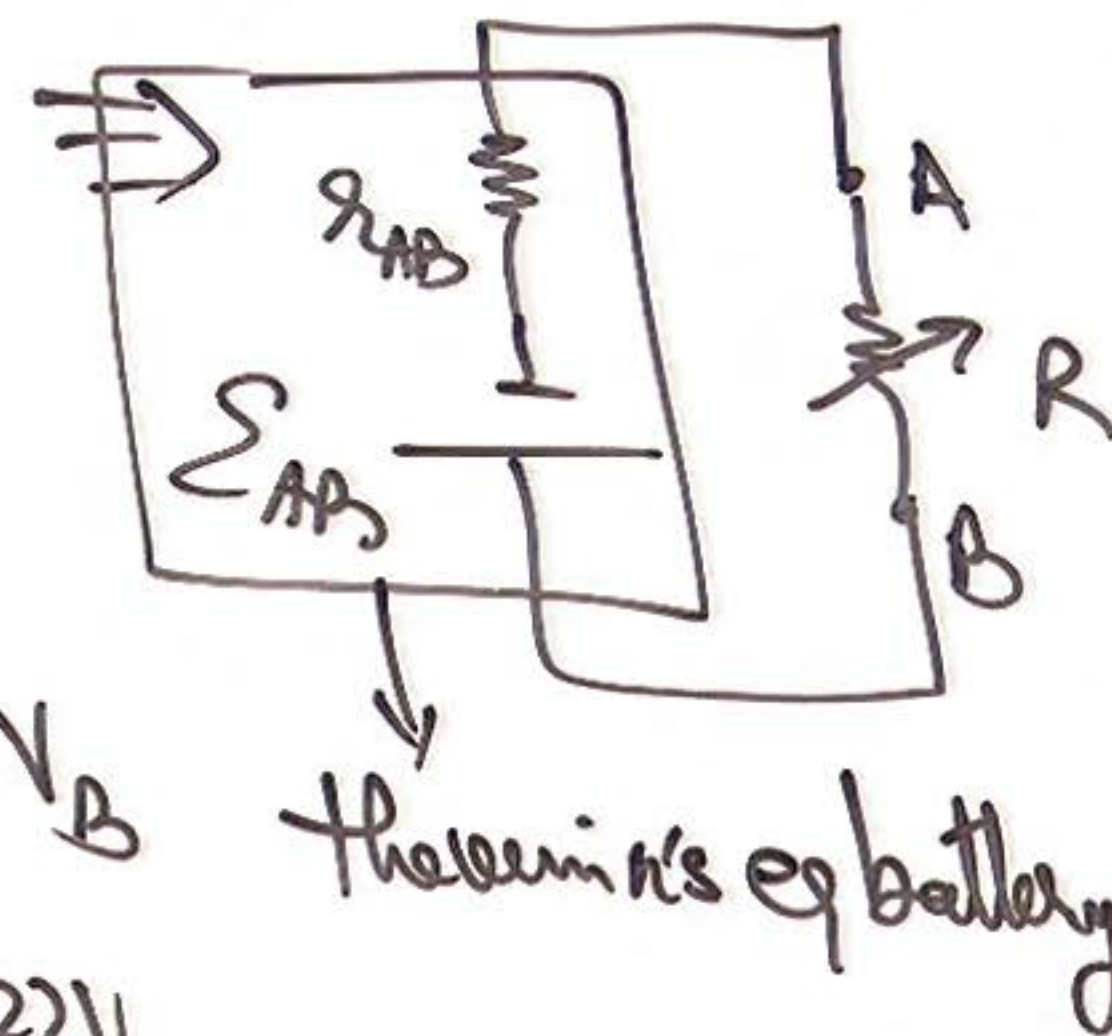
$$i = \frac{18}{6} = 3 \text{ A}$$

$$V_A + 10 + 12 = V_B$$

$$V_B - V_A = 22 \text{ V}$$

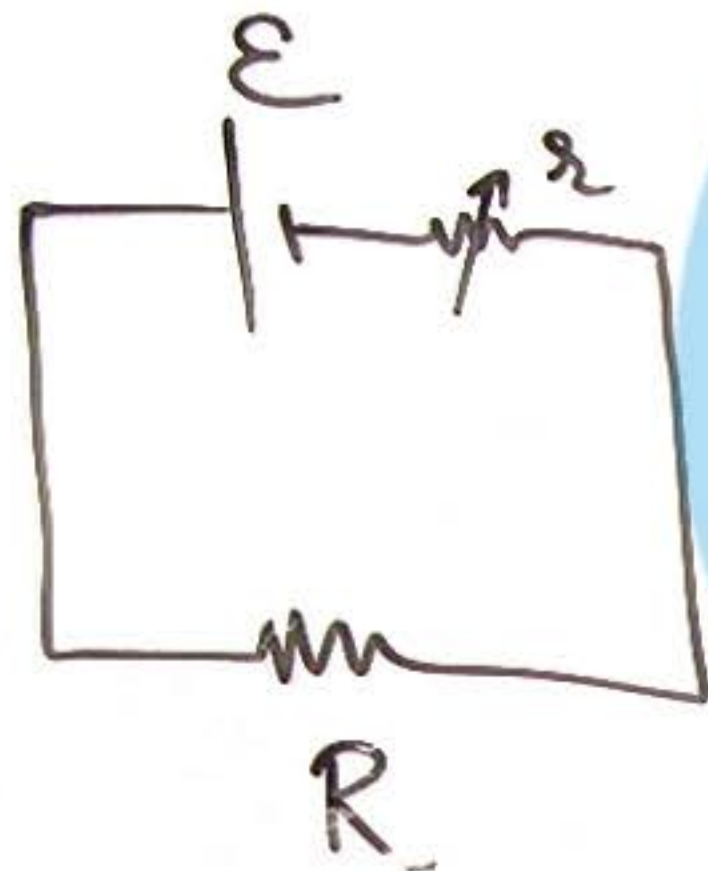
$$r_{AB} = \frac{4}{3} \Omega$$

$$\mathcal{E}_{AB} = 22 \text{ V}$$

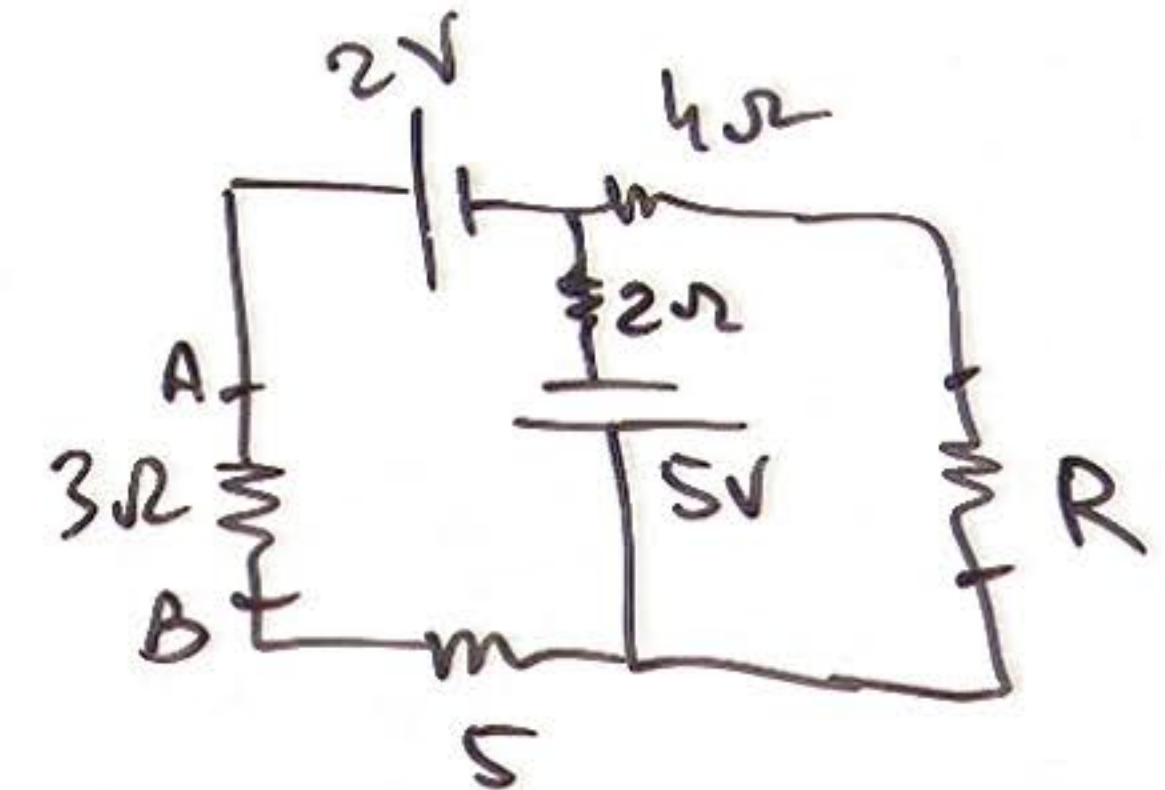


⑦ Power Transfer in case of
Variable internal resistance:

$$P_{\max} = \frac{\mathcal{E}^2}{R}$$



If ext R is Constant
then power supplied to it
will be max when r is min
 $\Rightarrow r = 0$

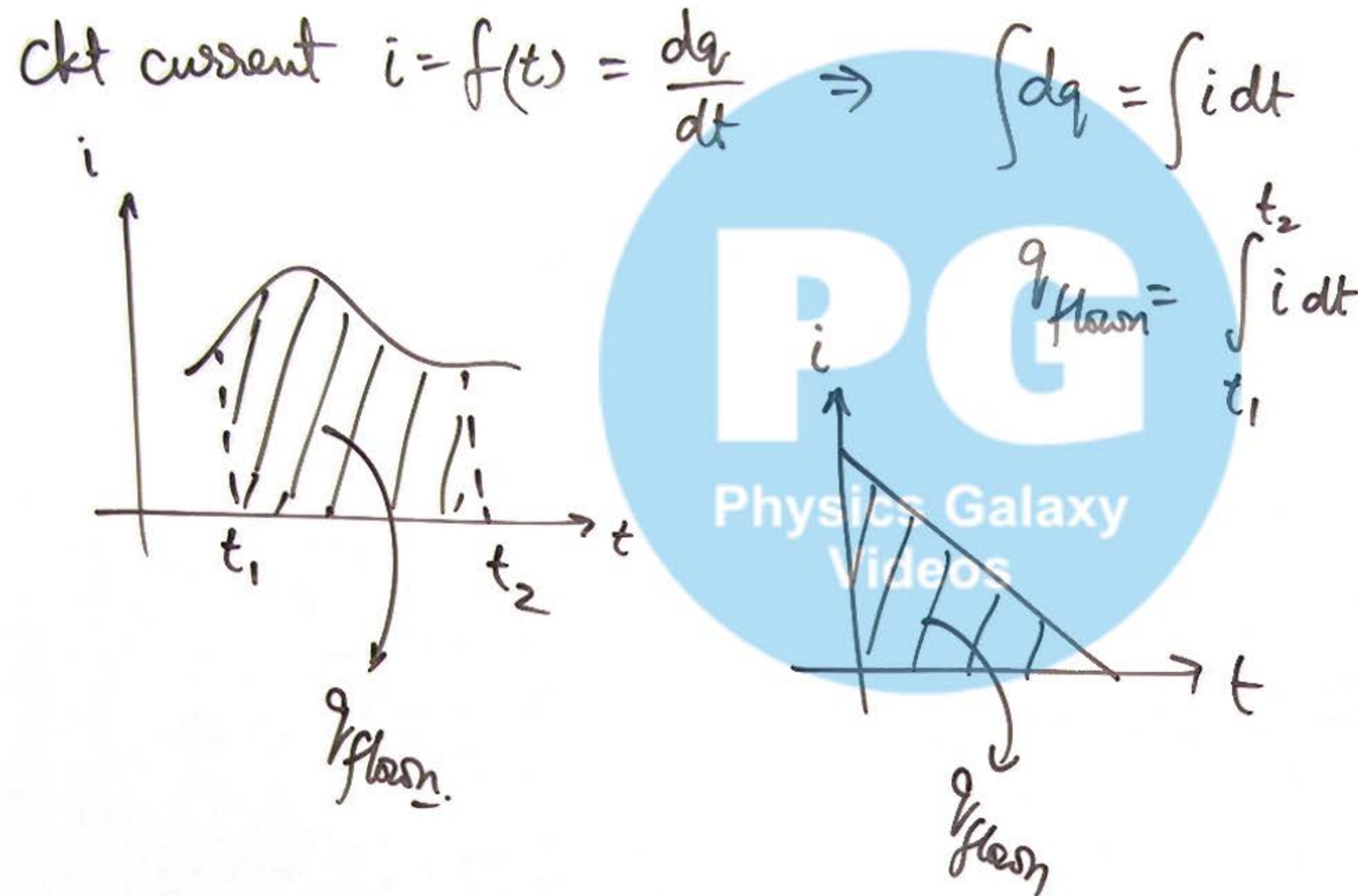


$$R_i = 5 + \frac{2(4+R)}{2+4+R}$$

R_i to be min

$$\underline{\underline{R = 2}}$$

⑧ Charge flown through a ckt:



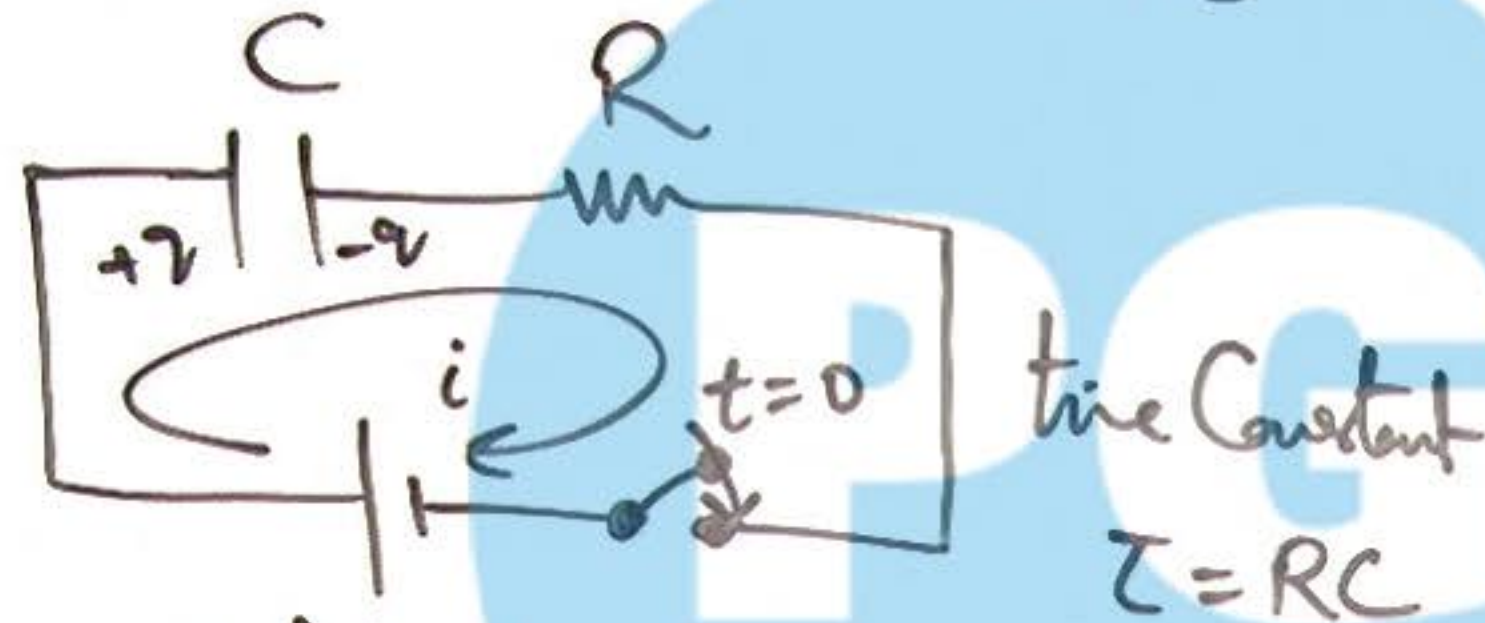
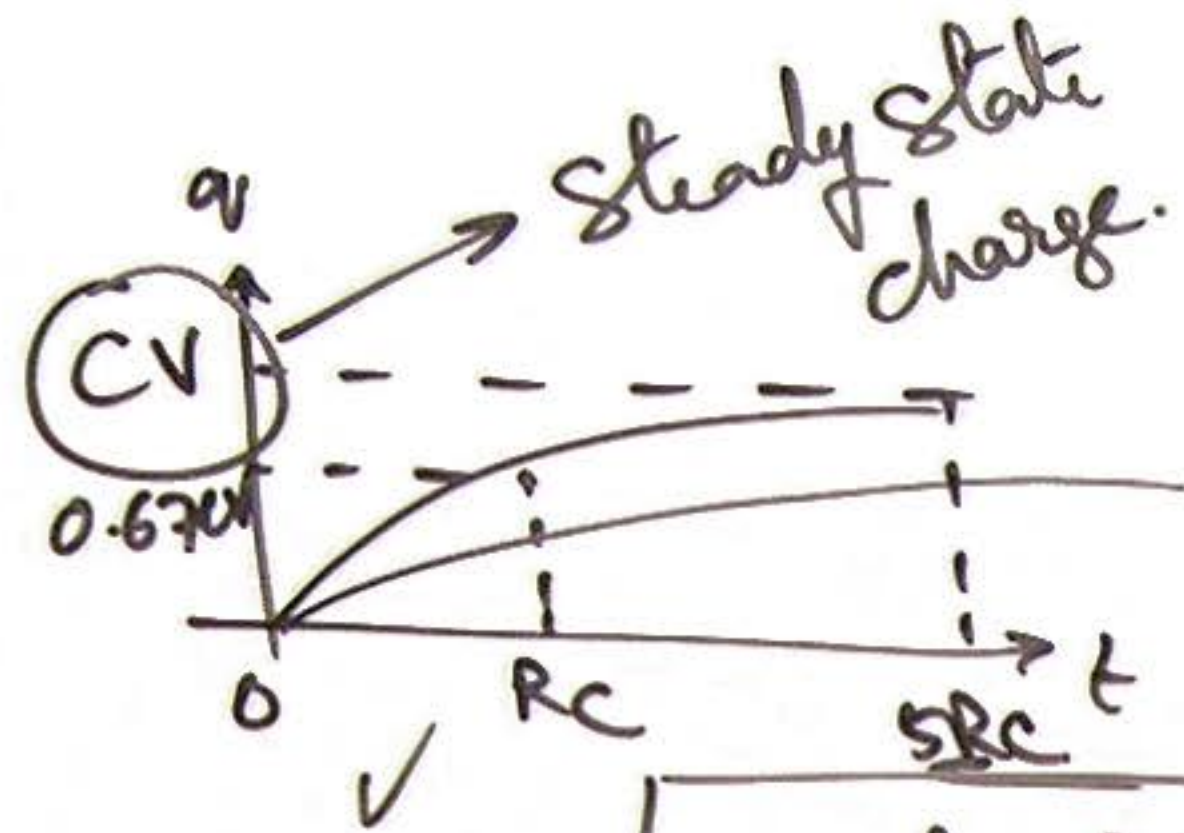
⑨ Heat produced in a Resistance due to time Varying Current:

$$H = \int_{t_1}^{t_2} i^2 R dt$$

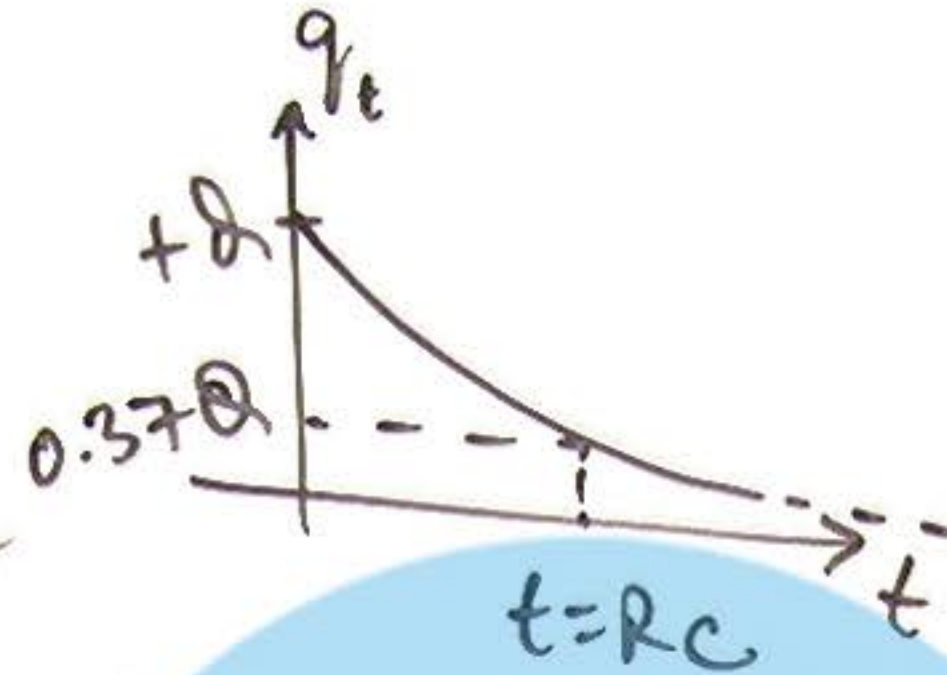
$f(t)$

10) Charging & Discharging of a Capacitor :

(Transient RC Ckt)



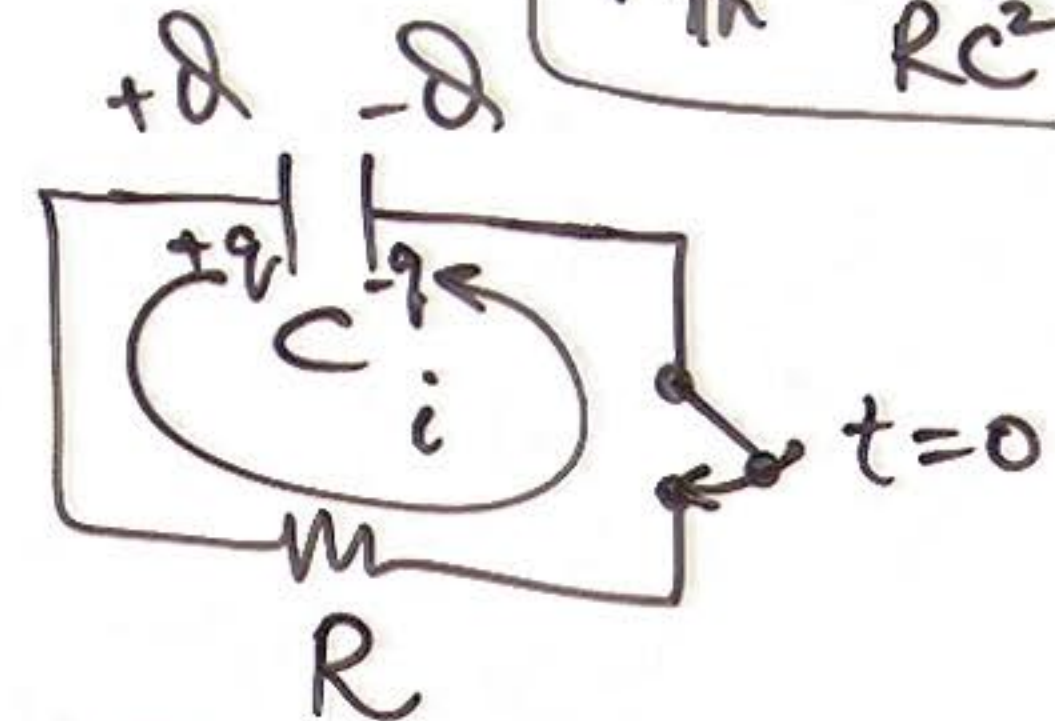
$U = \frac{1}{2} CV^2 (1 - e^{-t/RC})^2$	$q_t = CV(1 - e^{-t/RC})$
$P_{th} = \frac{V^2}{R} e^{-2t/RC}$	$i_t = \frac{V}{R} e^{-t/RC}$



$$V_t = \frac{Q}{C} e^{-t/RC}$$

$$U_t = \frac{Q^2}{2C} e^{-2t/RC}$$

$$P_{th} = \frac{Q^2}{RC^2} e^{-2t/RC}$$

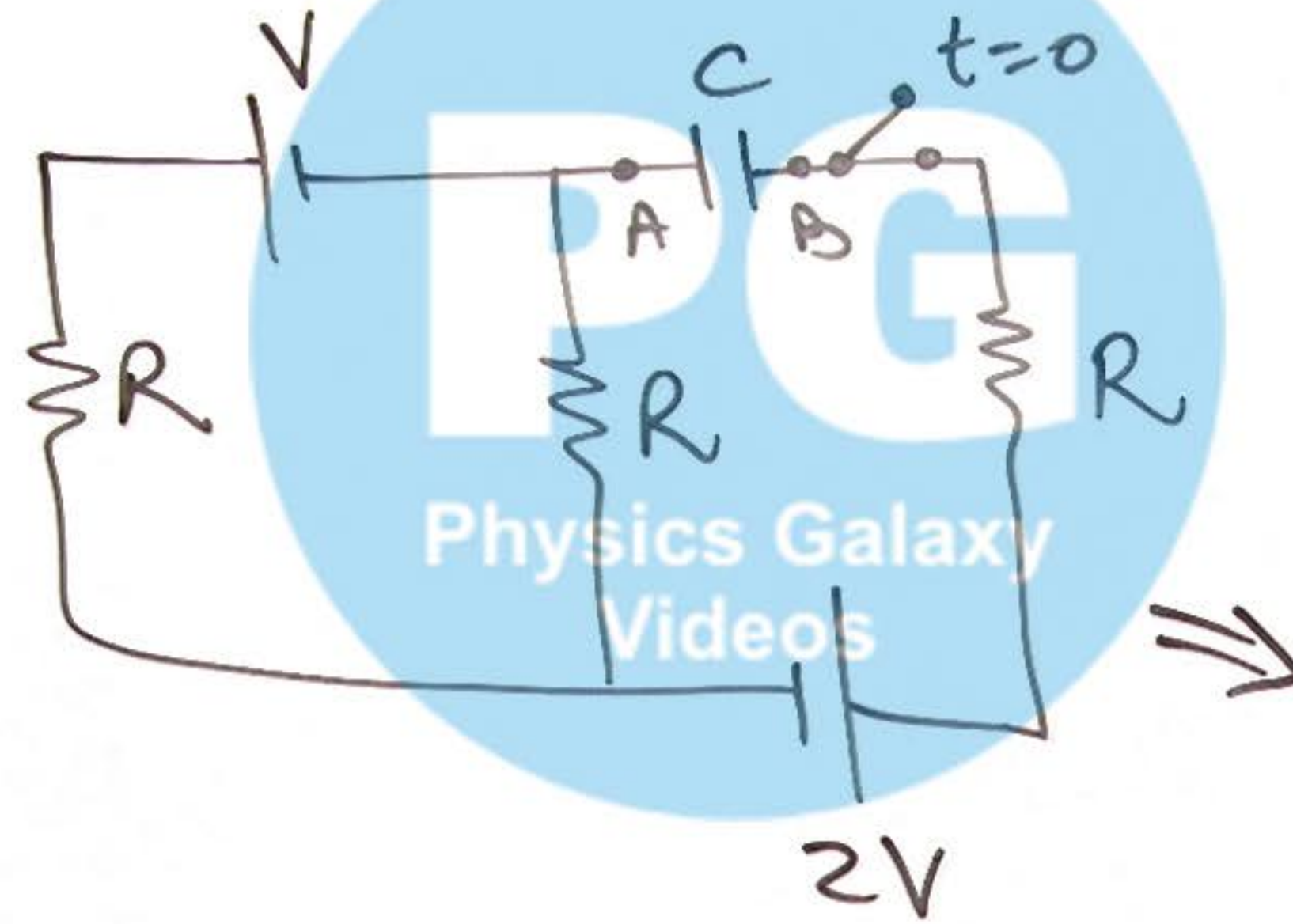


$$q_t = Q e^{-t/RC}$$

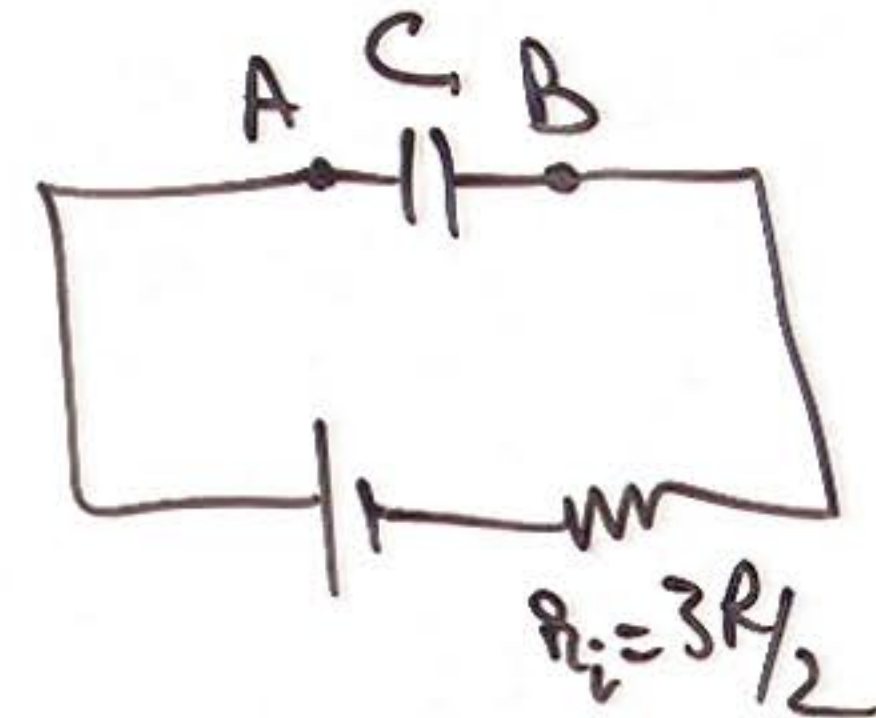
$$i_t = -\frac{Q}{RC} e^{-t/RC}$$

⑪ Time Constant of any RC ckt:

Ex: Find time constant of given ckt.



$$\tau = \frac{3R}{2}C \quad \checkmark$$

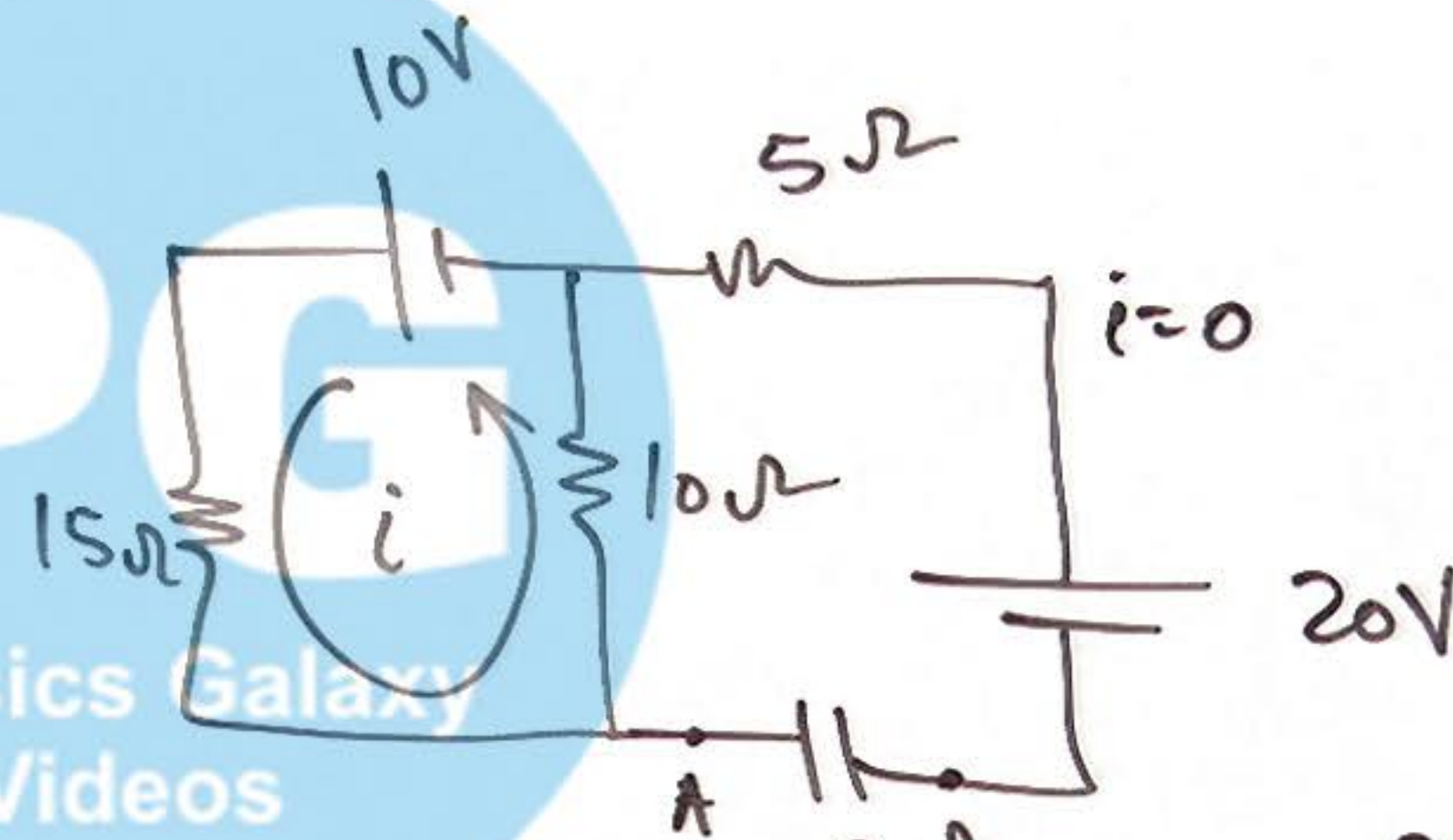


12 Steady State Analysis of RC Circuits:

$$i = 0 \checkmark$$

in any branch of Capacitor.

IL: find charge on capacitor plates in steady state



$$i = \frac{10}{25} = \frac{2}{5} A = 0.4 A$$

$$V_A - 4 - 20 = V_B \Rightarrow V_A - V_B = 24V$$

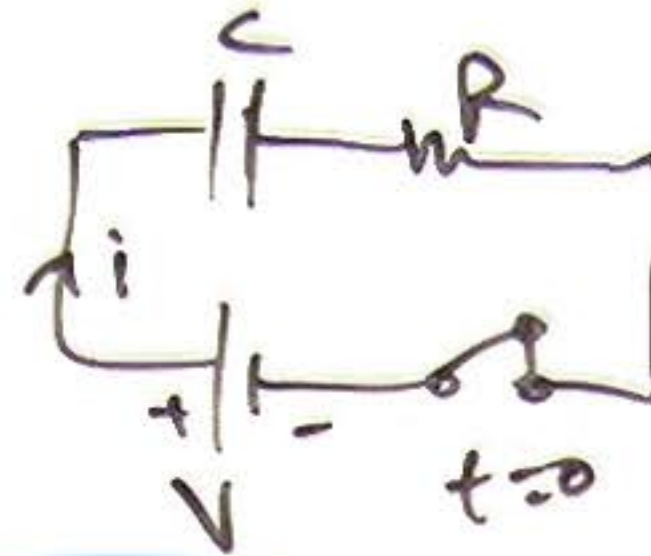
$$q_c = C(V_A - V_B)$$

$$= 5 \times 24 \mu C$$

$$= 120 \mu C \text{ Ans} \checkmark$$

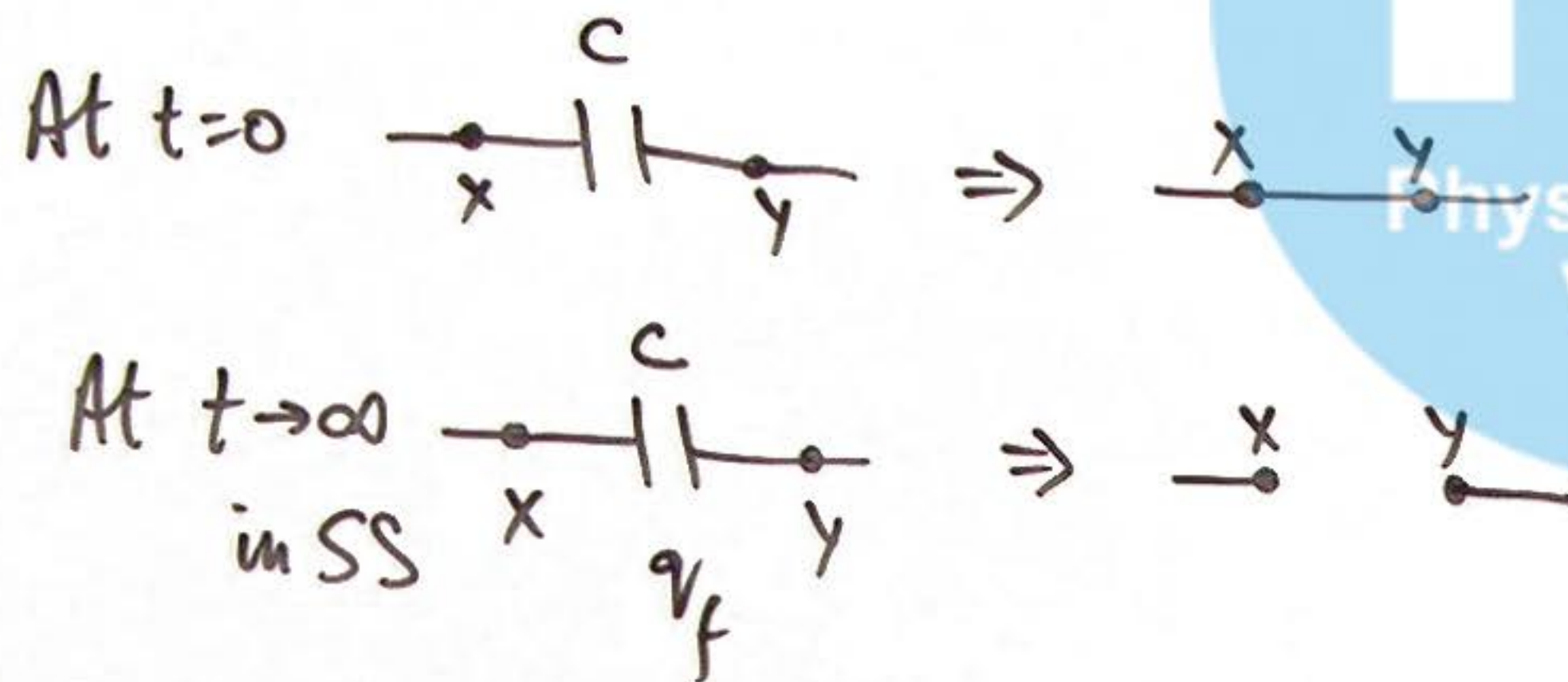
13 Initial & Steady State

Behavior of Capacitors in RC ckt:

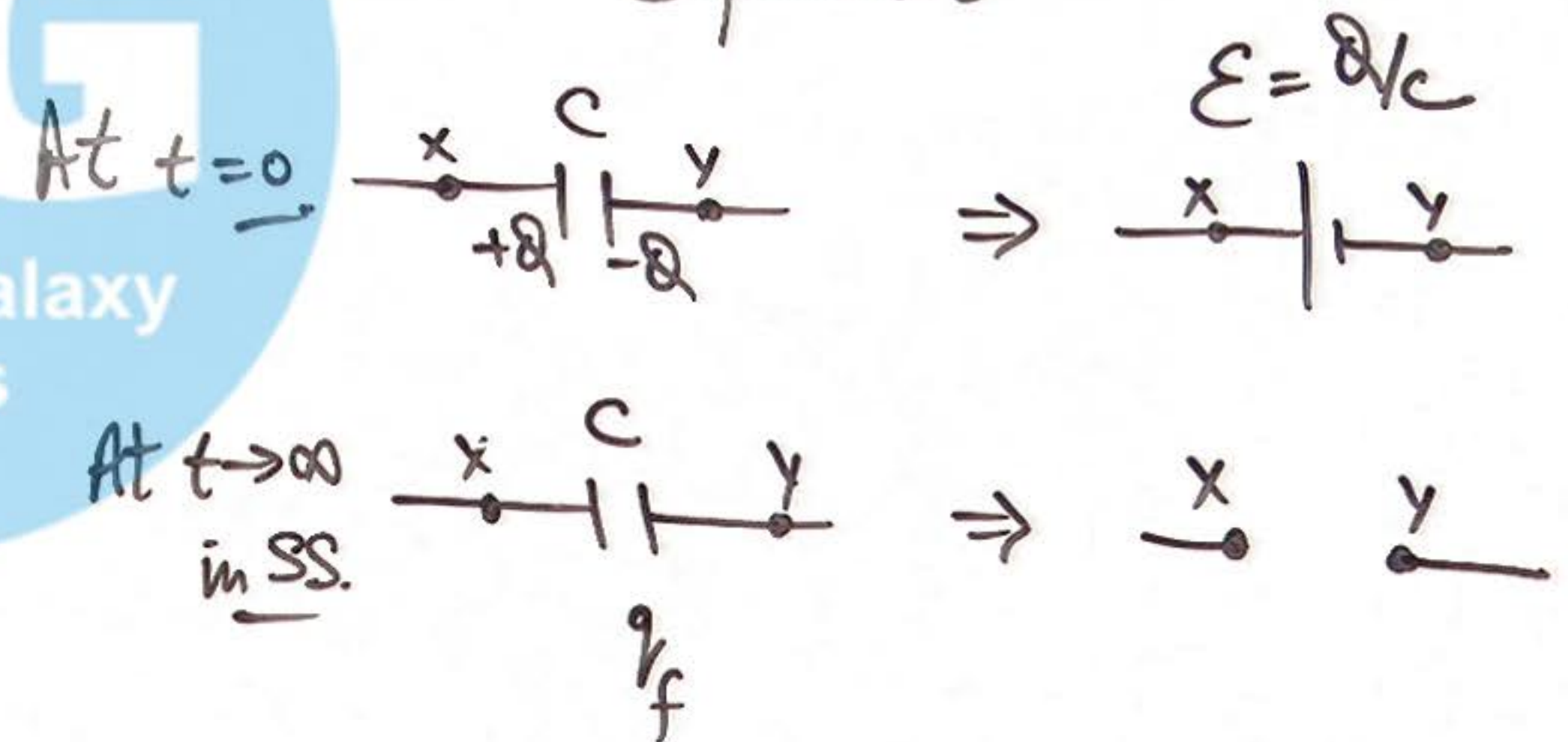


$$i = \frac{V}{R} e^{-t/RC} \rightarrow t=0 \quad i = \frac{V}{R}$$

Initially Uncharged
Capacitor



Initially Charged
Capacitor



PG
Physics Galaxy
Videos

14 Leaky Capacitor:

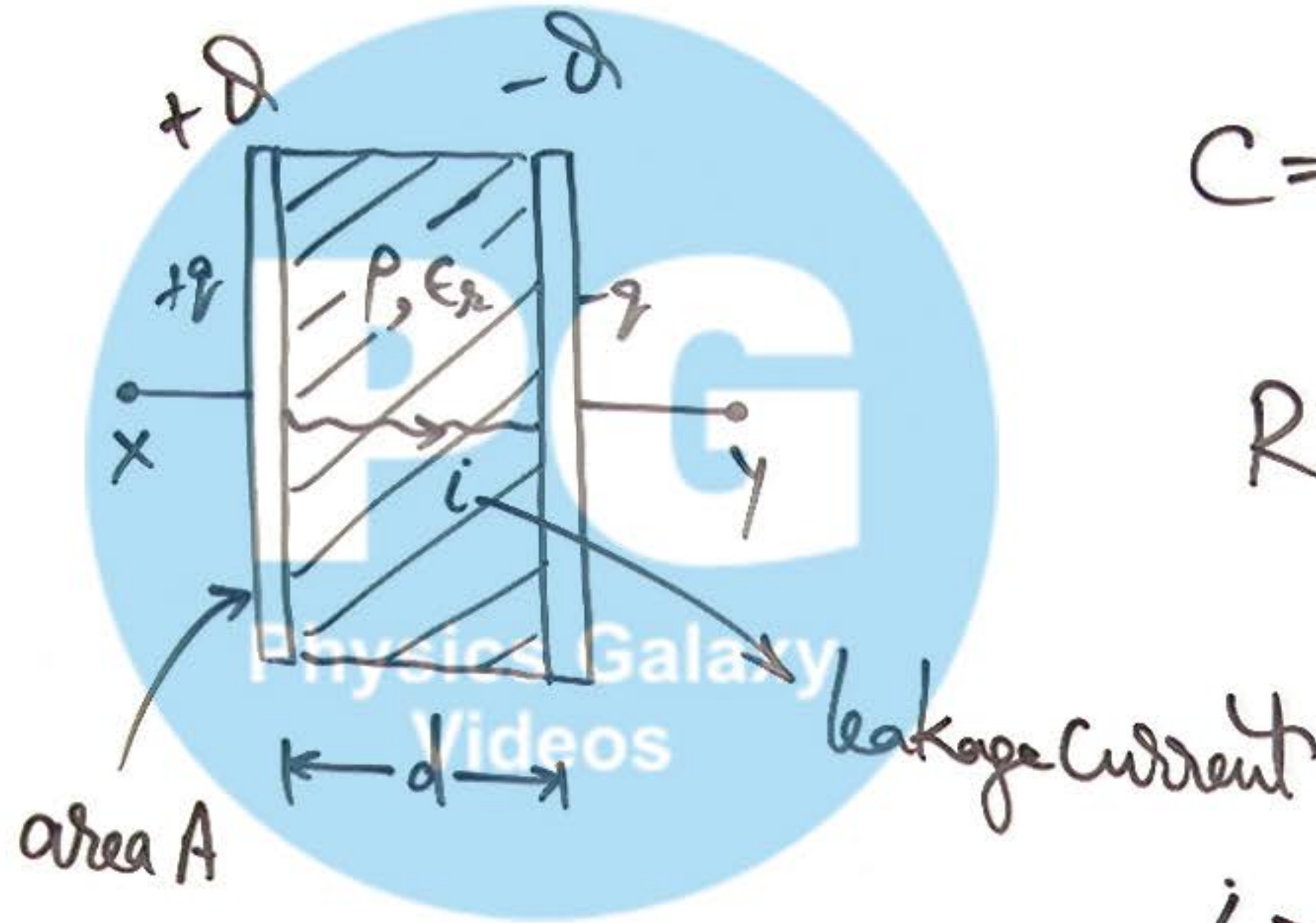
time Constant for discharging of Capacitor

$$\tau = RC$$

$$\tau = \rho \epsilon_r \epsilon_0$$

independent of A & d. (dimension of c)

$$q_t = Q e^{-t/\rho \epsilon_0 \epsilon_r}$$



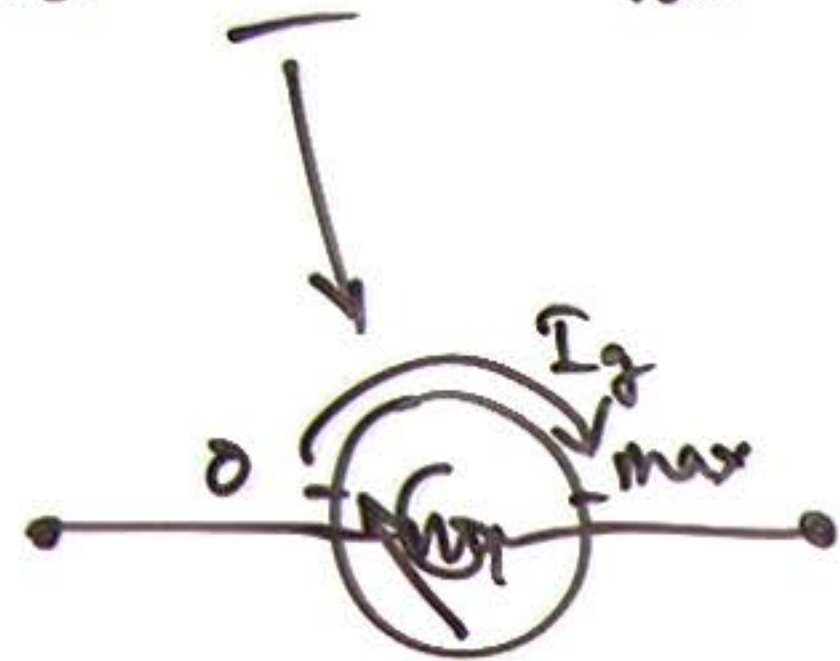
$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$R = \frac{\rho d}{A}$$

$$i_t = \frac{Q}{\rho \epsilon_0 \epsilon_r} e^{-t/\rho \epsilon_0 \epsilon_r}$$

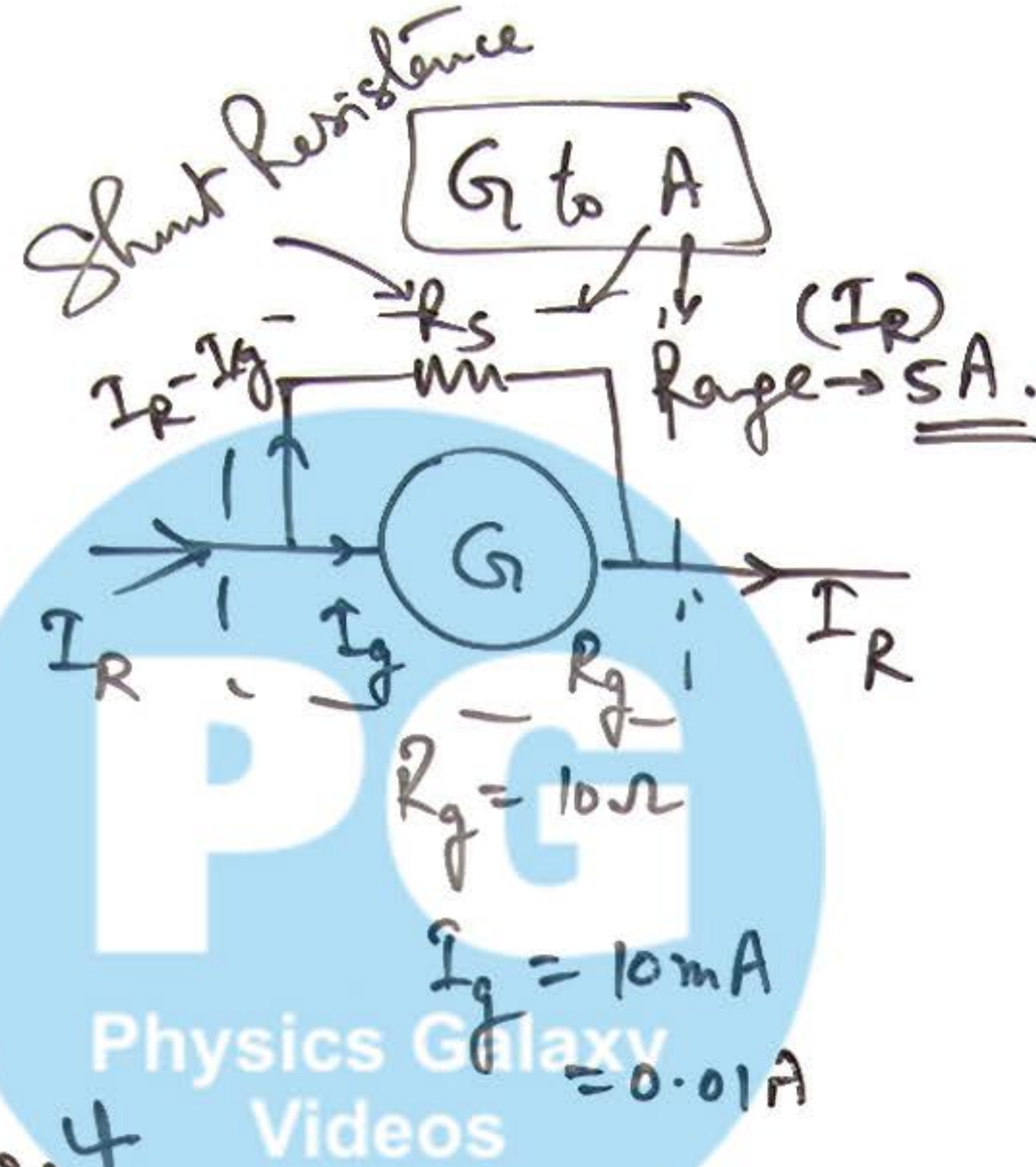
15 Deflection Type Meters:

Galvanometer Voltmeter Ammeter



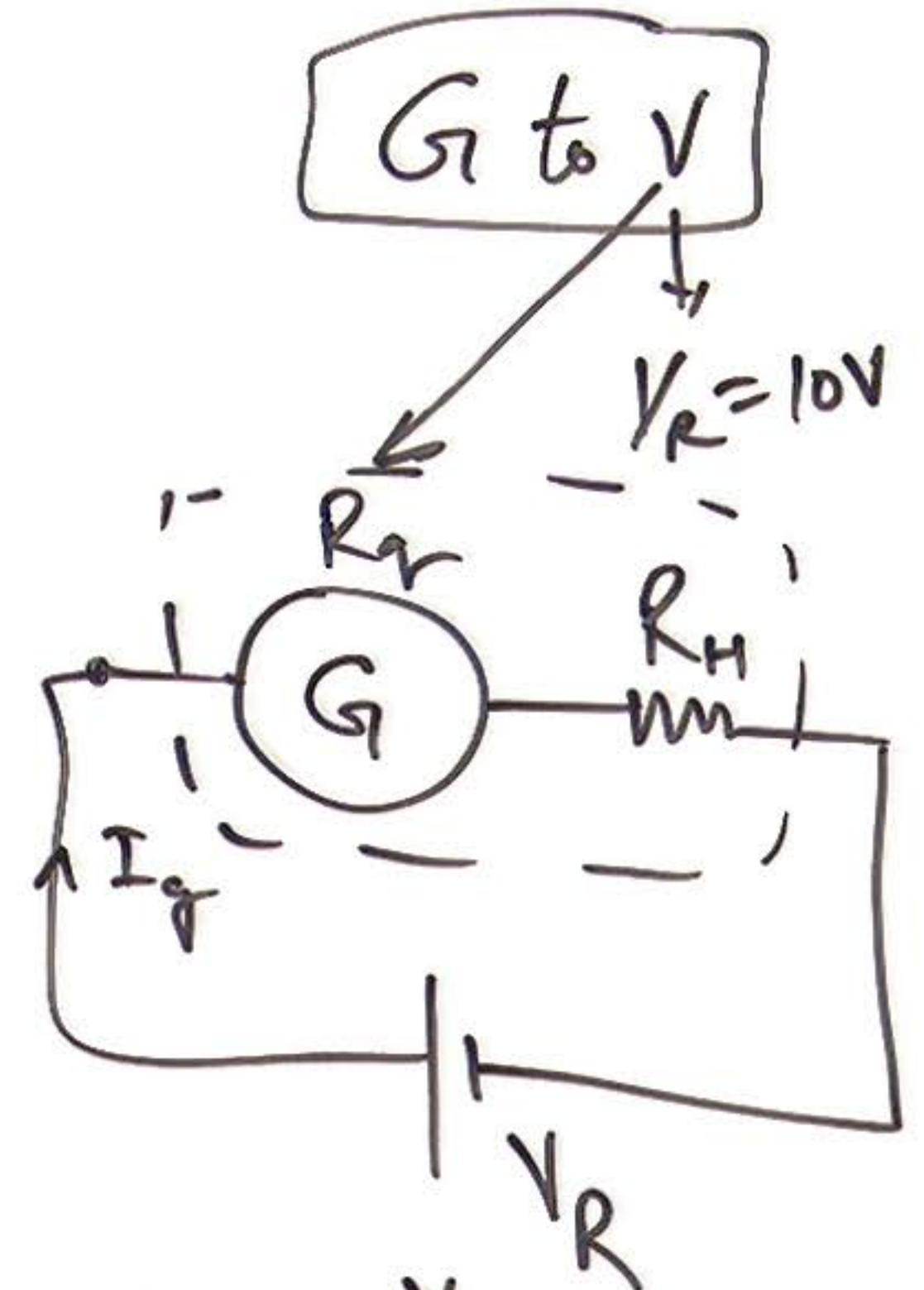
$0-10\Omega$ $R_g \rightarrow$ Coil Resistance

$I_g \rightarrow$ Full deflection Current
mA



$$(I_R - I_g)R_s = I_g R_g$$

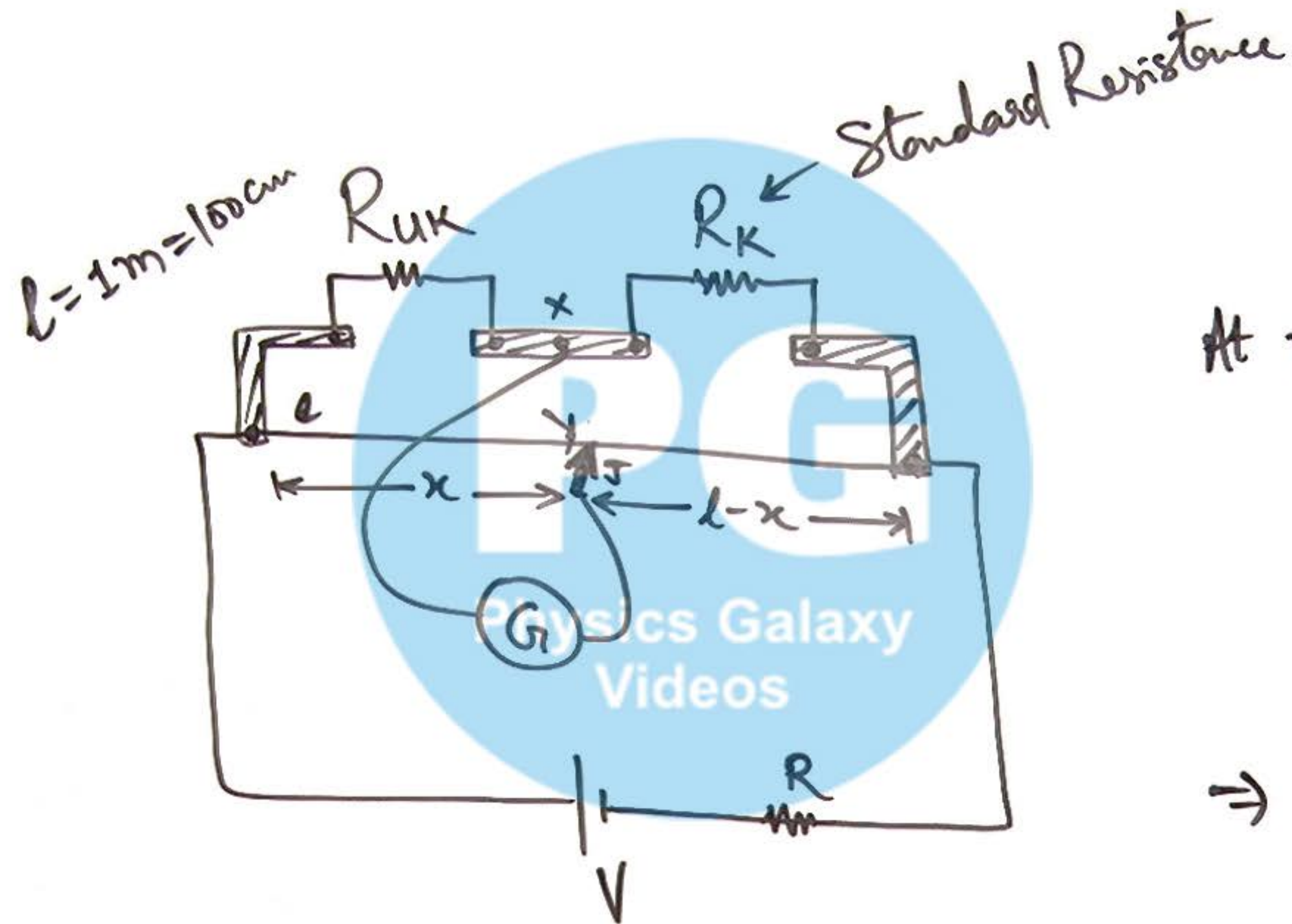
$$\Rightarrow R_s = \frac{I_g R_g}{I_R - I_g}$$



$$I_g = \frac{V_R}{R_g + R_H}$$

$$\Rightarrow R_H = \frac{V_R}{I_g} - R_g$$

16 Meter Bridge:



At null deflection, we use

$$\frac{R_{LK}}{x} = \frac{R_K}{l-x}$$

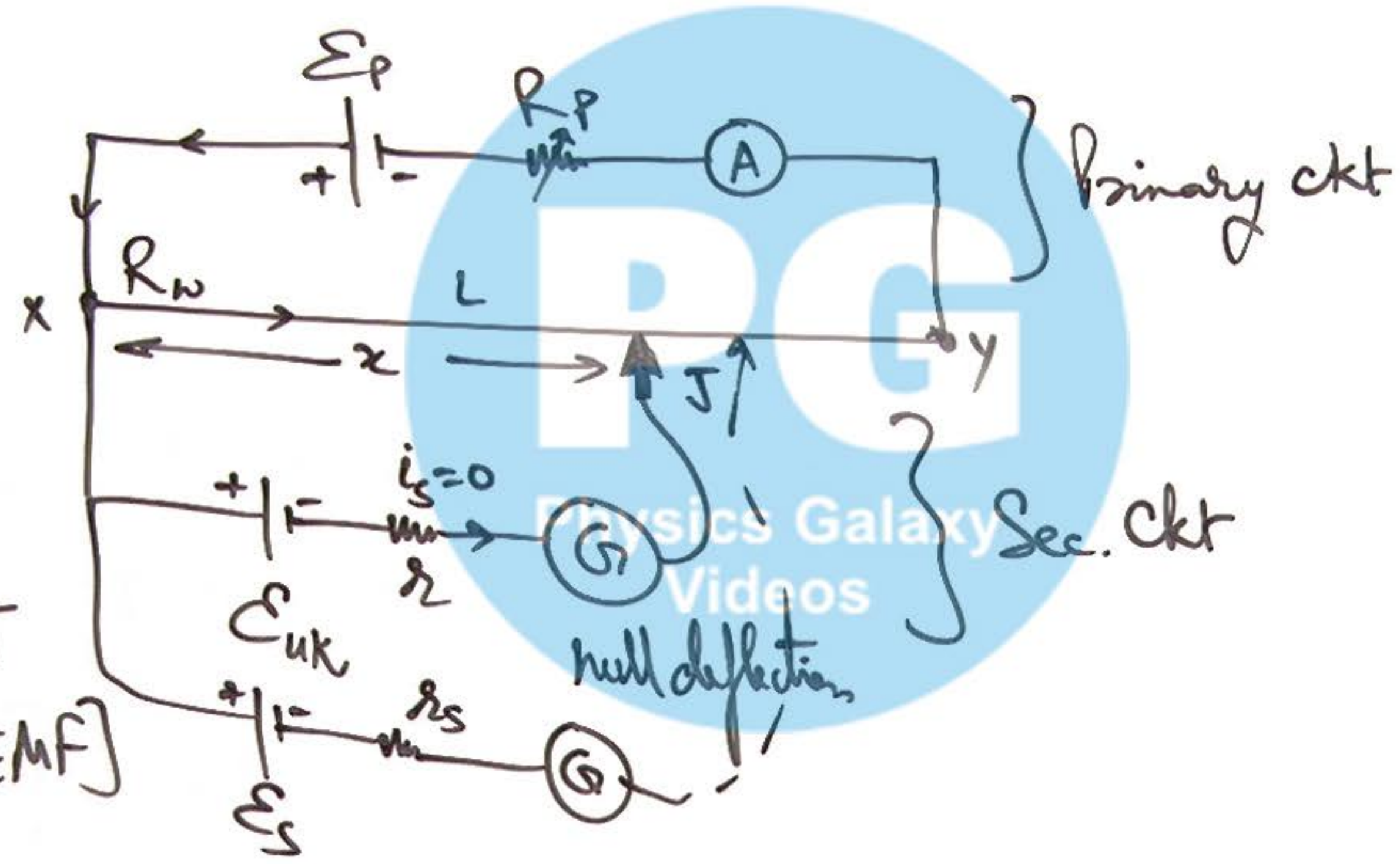
$$\Rightarrow R_{LK} = \left(\frac{R_K x}{100-x} \right)$$

①7 Potentiometer :

used to measure unknown EMFs:

Accuracy is high because $i=0$ at the point of measurement of P.D. [EMF]

$$i_p = \frac{\mathcal{E}_p}{R_w + R_p} \Rightarrow V_{xy} = i_p \cdot R_w \Rightarrow \text{potential gradient } \lambda = \frac{V_{xy}}{L} \text{ Volt/cm.}$$



At null deflection

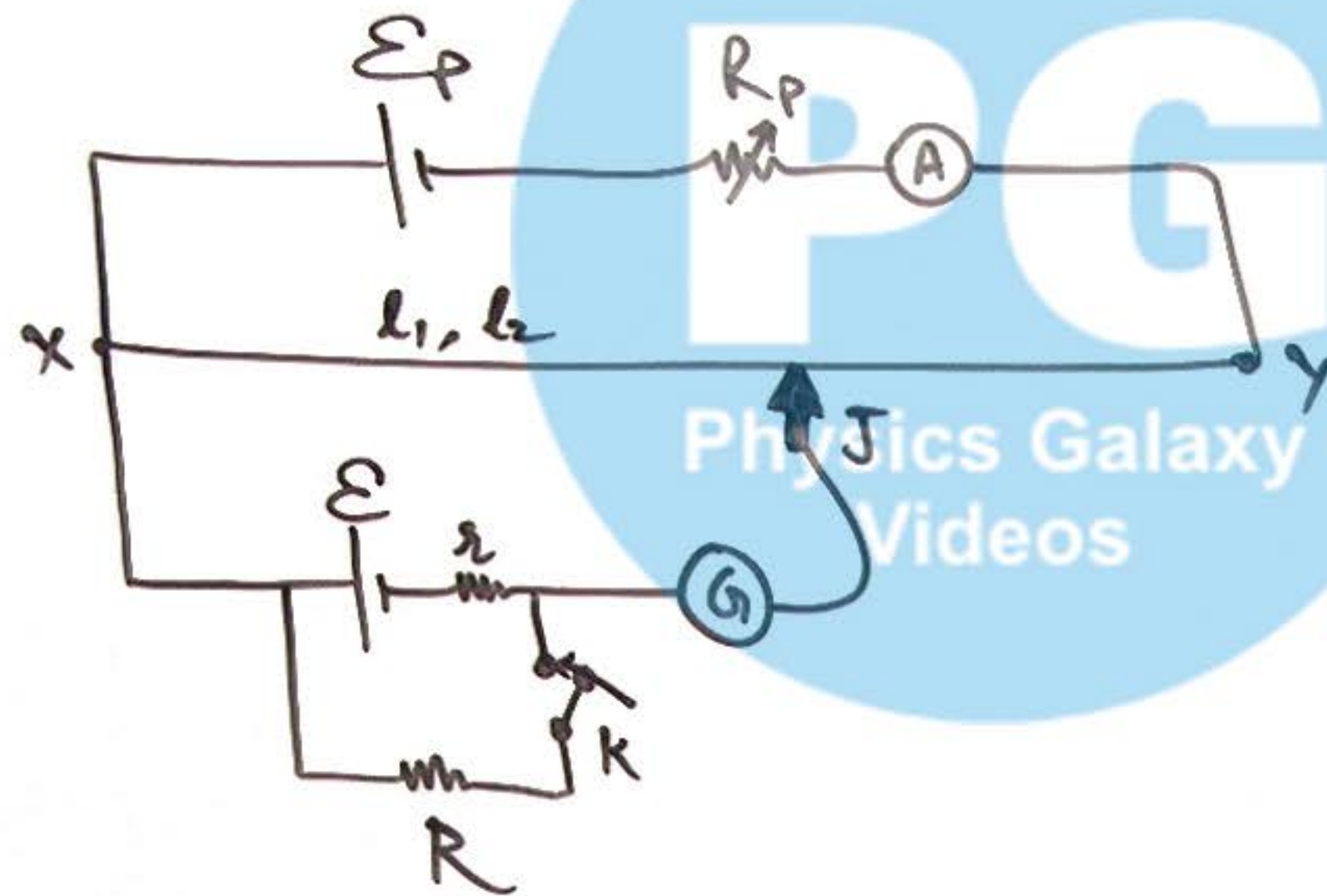
$$\mathcal{E}_{uk} = \lambda x$$

If x_1 and x_2 are balancing lengths for cells $\mathcal{E}_{uk}$ & $\mathcal{E}_s$

$$\Rightarrow \mathcal{E}_s = \lambda x_s$$

$$\Rightarrow \mathcal{E}_{uk} = \frac{x}{x_s} \mathcal{E}_s \checkmark$$

⑮ Internal Resistance Measurement of a Cell by a Potentiometer :



Balancing length

$K \text{ open} \rightarrow l_1 \Rightarrow \mathcal{E} = \lambda l_1$

$K \text{ closed} \rightarrow l_2 \Rightarrow \frac{\mathcal{E}R}{r+R} = \lambda l_2$

$$\Rightarrow \boxed{r = \left(\frac{l_1 - l_2}{l_2} \right) R}$$

19) Post Office Box:

